



School of Future Tech

Case Study Report

on

Case Study 39: Coppervale Semiconductor Fab Network Outage Investigation: Capstone Diagnostics

(STP · OSPF · EIGRP · BGP · ICMP, simulated in Cisco Packet Tracer)

by

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Source: github.com/Yashhh710/coppervale-network-outage-diagnostics

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1. Introduction to the Case Study

Coppervale Semiconductor Fab runs three sites: a Head Office (HQ), a Wafer Fabrication Plant that still uses a legacy EIGRP network, and a Test & Packaging Facility. They are tied together by an OSPF backbone, and the HQ has dual-ISP Internet access using BGP. For 48 hours, staff at every site reported intermittent problems. The network team captured a set of symptoms but had not found the root causes.

In this case study the author takes the role of the on-call network engineer. The five symptoms (A to E) are reproduced in a Cisco Packet Tracer lab, each is diagnosed with Cisco IOS show commands, a fix is applied, and a verification step proves the fix worked. The five faults span Layer 2 switching (STP Root Guard), the OSPF WAN, EIGRP, BGP and ICMP traffic analysis.

The report follows one repeatable workflow for every fault: Reproduce, Observe, Explain, Fix, Verify. It ends with a root-cause table, a written narrative on whether the faults are related, and a prioritized remediation plan ranked by business impact.

2. Problem Statement / Case Background (Abstract)

Problem Statement

Coppervale's network team captured symptoms at several sites but found no root cause. As the on-call engineer, diagnose each symptom, identify the most likely root cause, and propose a fix and a verification step for each.

Symptom Log

Sym.	Location	Observed behaviour
A	Fab Plant access layer	A switch port configured with Root Guard enters a root-inconsistent (blocking) state. Logs show the port received a superior BPDU from a neighboring switch whose priority was recently, and apparently mistakenly, lowered below the intended root bridge's.
B	HQ–Fab Plant OSPF link (multipoint)	On a multipoint WAN segment, one router's OSPF neighbor entry for a specific remote router is stuck in the ATTEMPT state and never progresses, while its other neighbors on the same segment are fully adjacent.
C	Fab Plant–Test & Packaging EIGRP route	A router recently configured as an EIGRP stub is not advertising a route that the rest of the network expects it to provide, even though the route exists in its local routing table.
D	HQ Internet edge (BGP)	The eBGP session to the secondary ISP repeatedly resets. Logs show the session was torn down immediately after the router logged a message about exceeding a configured maximum-prefix limit.
E	Fab Plant access layer (Wireshark)	A packet capture shows a burst of unusually large, fragmented ICMP echo request packets arriving in rapid succession, all targeting a single internal host.

Objectives

- Symptom A: explain what Root Guard prevents, why a superior BPDU triggers the root-inconsistent state, and which commands confirm the cause and restore the port.
- Symptom B: explain why an OSPF neighbor on a multipoint (NBMA-style) segment can stay in ATTEMPT while other neighbors become FULL, and state the fix.
- Symptom C: explain the difference between EIGRP stub types, why an incorrect type silently withholds a route, and the fix.
- Symptom D: explain what the BGP maximum-prefix limit is for, why exceeding it tears the session down, how to check the limit and the received routes, and how to re-establish the session safely.
- Symptom E: state what the capture pattern indicates, its likely intent, and two concrete mitigations.
- Decide whether the five symptoms share one common trigger or more than one unrelated fault, justified with evidence.
- Produce a single root-cause report ranking the symptoms by business impact, with an order of remediation and a verification check for each.

Outcomes and Deliverables

Outcomes: diagnose the symptoms systematically using networking and security concepts; identify likely root causes and propose corrective actions in priority order; define verification steps that show whether each fix resolved the symptom.

Deliverable	Where in this report
1. Root-cause analysis table (Symptom, Cause, Evidence, Fix, Verification)	Section 6.1
2. Short narrative (300-500 words): are the symptoms related?	Section 6.2
3. Prioritized remediation plan with business-impact justification	Section 6.3
4. Annotated before / after command output for each fix	Sections 5.2 to 5.6

Abstract

The project recreates Coppervale's network in Cisco Packet Tracer with three routers (R1, R2, R3), a simulated secondary ISP (ISP2), four switches, two PCs and a server. A healthy baseline is built first. Each of the five faults is then introduced one at a time with a small, controlled configuration change and diagnosed from show-command evidence. The causes are: a switch priority set to 0 that triggers Root Guard (A), an OSPF network-type mismatch on the NBMA WAN (B), an EIGRP stub set to receive-only (C), a BGP maximum-prefix limit of 3 against 5 advertised prefixes (D), and a large fragmented ICMP flood from one host (E). The analysis concludes that the symptoms come from more than one unrelated fault, with A

and C possibly sharing a change window. Remediation is ranked B, C, A, E, D by business impact.

3. Case Study Design

3.1 Lab Topology

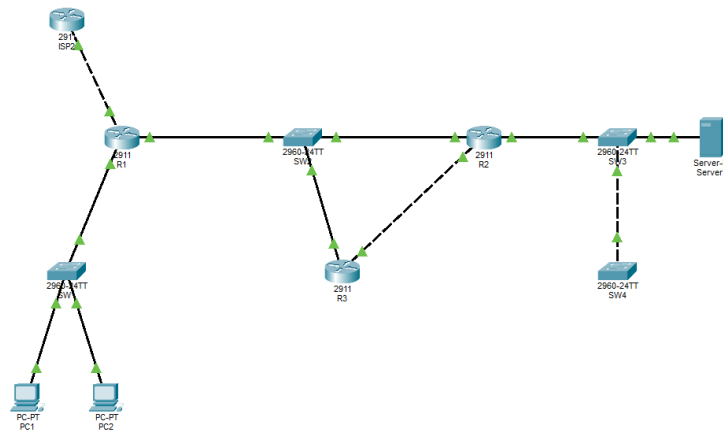


Figure 1: Packet Tracer topology of the Coppervale network (screenshot from the project)

The topology joins the HQ LAN (R1, SW1, PC1, PC2), a multipoint WAN segment (SW2 with R1, R2 and R3), the Fab Plant (R2, SW3, SW4, Server1), the Test & Packaging site (R3) and the secondary ISP (ISP2). The dashed link between R2 and R3 is the EIGRP link. ISP1 is optional in the case design and is left out of the lab because a 2911 router has only three gigabit ports; symptom D is tested on ISP2, the secondary ISP, as in the case study.

3.2 Devices and Roles

Device	Type	Role
R1	Cisco 2911	HQ router: OSPF hub (priority 255) and eBGP edge to ISP2 (AS 65100). Location of symptom D and hub for B.
R2	Cisco 2911	Fab router: OSPF toward the WAN, EIGRP toward R3, redistributes between them.
R3	Cisco 2911	Test & Packaging router: EIGRP stub (symptom C), also attached to the WAN switch.
ISP2	Cisco 2911	Secondary ISP, AS 65002. Advertises five test networks through loopbacks.
SW1	Cisco 2960	HQ LAN switch (R1, PC1, PC2).
SW2	Cisco 2960	Multipoint WAN segment where R1, R2 and R3 meet (symptom B). Default configuration.
SW3	Cisco 2960	Fab access switch and intended STP root; Root Guard on Fa0/3 (symptom A).

Device	Type	Role
SW4	Cisco 2960	Neighbor switch used to send the superior BPDU (symptom A).
PC1 / PC2	PC	PC1 is a normal HQ user. PC2 is the attacker or infected PC that sends the ping flood (E).
Server1	Server	Fab internal host and target of the ICMP flood.

3.3 IP Addressing

Device	Interface	Address	Purpose
R1	Gi0/0	192.168.10.1/24	HQ LAN
R1	Gi0/1	10.0.0.1/24	WAN (OSPF NBMA)
R1	Gi0/2	172.16.2.1/30	Link to ISP2
R2	Gi0/0	10.0.0.2/24	WAN
R2	Gi0/1	192.168.20.1/24	Fab LAN
R2	Gi0/2	10.2.0.1/30	EIGRP link to R3
R3	Gi0/0	10.0.0.3/24	WAN
R3	Gi0/1	10.2.0.2/30	EIGRP link to R2
R3	Loopback0	192.168.30.1/24	Test & Packaging LAN
ISP2	Gi0/0	172.16.2.2/30	Link to R1
ISP2	Lo1 to Lo5	198.51.100.1 to 198.51.104.1 /24	Five advertised test networks
PC1	NIC	192.168.10.10/24, GW 192.168.10.1	Normal HQ client
PC2	NIC	192.168.10.66/24, GW 192.168.10.1	Flood source
Server1	NIC	192.168.20.100/24, GW 192.168.20.1	Flood target

3.4 Protocol Design

Protocol	Role in the network
STP	Prevents Layer 2 loops by electing one root bridge (lowest priority wins) and blocking redundant paths. SW3 is the intended root (priority 4096) with Root Guard on Fa0/3; SW4 uses the normal 32768.
OSPF	Backbone between R1, R2 and R3 on 10.0.0.0/24 in area 0. The WAN is set as non-broadcast (NBMA), so neighbors are configured by hand on the hub. R1 is the hub (priority 255); R2 and R3 are spokes (priority 0).
EIGRP	AS 100 between R2 and R3. R2 redistributes between OSPF and EIGRP; R3 is a stub (connected summary) that advertises 192.168.30.0/24.
BGP	eBGP between R1 (AS 65100) and ISP2 (AS 65002), with a maximum-prefix limit on R1 to protect it from excess routes.
ICMP	Echo traffic is used for connectivity testing and to reproduce the ping flood.

3.5 Healthy Baseline Configuration

The working state is built first, so every fault can be compared with a known-good network.
Key settings:

```
R1: interface gi0/1 / ip ospf network non-broadcast / ip ospf priority 255
    router ospf 1 / neighbor 10.0.0.2 / neighbor 10.0.0.3
R2: interface gi0/0 / ip ospf network non-broadcast / ip ospf priority 0
R3: interface gi0/0 / ip ospf network non-broadcast / ip ospf priority 0
R3: router eigrp 100 / eigrp stub connected summary
R1: router bgp 65100 / neighbor 172.16.2.2 remote-as 65002
    neighbor 172.16.2.2 maximum-prefix 10
SW3: spanning-tree vlan 1 priority 4096 ; interface fa0/3 / spanning-tree guard root
SW4: spanning-tree vlan 1 priority 32768
```

Baseline checks: R1 shows R2 and R3 as FULL; R1 has routes to 192.168.20.0 and 192.168.30.0; ISP2 shows 5 prefixes; SW3 has no inconsistent ports; PC1 can ping 192.168.20.100 and 192.168.30.1.

4. Methods, Algorithms & Technology Applied

4.1 Troubleshooting Methodology

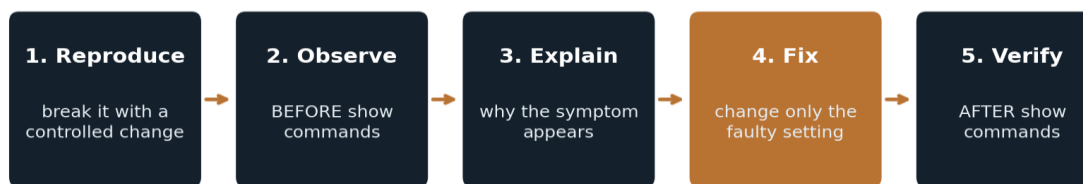


Figure 2: Five-step workflow applied to every symptom

Faults are introduced and fixed one at a time. For each, the same show command is run before and after the change, only the setting that caused the fault is changed, and security protections such as Root Guard and the BGP prefix limit stay enabled.

4.2 Key Concepts Behind Each Fault

Fault	Concept
A: Root Guard	Root Guard stops an unauthorized or misconfigured switch from becoming the root bridge. The root decides the whole Layer 2 tree, so a new root would change traffic paths and could cause outages or let an attacker see traffic. When a Root Guard port receives a superior BPDU (lower bridge priority), the port is put into the root-inconsistent blocking state instead of accepting it. It recovers by itself when superior BPDUs stop.
B: OSPF NBMA	On a non-broadcast network OSPF cannot find neighbors with multicast, so the hub sends unicast hellos to each configured neighbor address. ATTEMPT means "hellos sent, nothing heard back". Both sides need matching NBMA settings. If the hub has no neighbor statement for a spoke, that spoke never appears at all.
C: EIGRP stub	A stub does not pass queries and advertises only what its type allows (see the table below). With receive-only the router learns routes but advertises none, not even its own connected LAN.

Fault	Concept
D: BGP max-prefix	The limit protects the router from receiving too many routes (an ISP mistake, a route leak or an attack) that could exhaust memory or CPU. The session is torn down instead of quietly dropping routes, because running with an incomplete, unpredictable routing table is worse than a loud failure.
E: ICMP flood	Echo requests larger than the 1500-byte MTU are fragmented. Many fragmented requests from one source to one host in a short time form a ping flood or Ping-of-Death style DoS that can exhaust CPU, reassembly memory or bandwidth.

EIGRP stub options (why a wrong type hides a route):

Stub option	What the stub advertises	Typical use
eigrp stub (default)	Connected and summary routes	Normal branch router
eigrp stub connected	Only connected networks	Branch with LANs only
eigrp stub summary	Only summary routes	Branch with summarized LANs
eigrp stub static / redistributed	Static or redistributed routes	Branch with static or redistributed routes
eigrp stub receive-only	Nothing; it only listens	Isolated router that must not share routes

4.3 Tools and Commands

Area	Commands
Interfaces / routing	show ip interface brief, show ip route, show ip protocols
STP	show spanning-tree, show spanning-tree inconsistentports, show spanning-tree interface fa0/3 detail
OSPF	show ip ospf neighbor, show ip ospf interface, show running-config section router ospf
EIGRP	show ip route eigrp, show ip protocols
BGP	show ip bgp summary, show ip bgp neighbors, show ip route bgp, clear ip bgp
ICMP / ACL	show access-lists; Wireshark filter icmp.type == 8 && ip.flags.mf == 1; Packet Tracer Simulation mode

4.4 Technology Stack

Item	Details
Simulator	Cisco Packet Tracer (project file coppervale_phase1.pkt)
Devices	Cisco 2911 routers, Cisco 2960 switches, PCs and a server
Operating system	Cisco IOS command line
Packet analysis	Packet Tracer Simulation mode; Wireshark filter shown as representative
Version control	Git and GitHub

5. Implementation Details and Snapshots

5.1 Baseline Interface Verification

The interface state of each router was captured with show ip interface brief and saved in the project's outputs folder. All used interfaces are up/up with the planned addresses, which confirms the baseline addressing.

```
R1# show ip interface brief
GigabitEthernet0/0 192.168.10.1 YES manual up up
GigabitEthernet0/1 10.0.0.1 YES manual up up
GigabitEthernet0/2 172.16.2.1 YES manual up up

R2# show ip interface brief
GigabitEthernet0/0 10.0.0.2 YES manual up up
GigabitEthernet0/1 192.168.20.1 YES manual up up
GigabitEthernet0/2 10.2.0.1 YES manual up up

R3# show ip interface brief
GigabitEthernet0/0 10.0.0.3 YES manual up up
GigabitEthernet0/1 10.2.0.2 YES manual up up
Loopback0 192.168.30.1 YES manual up up

ISP2# show ip interface brief
GigabitEthernet0/0 172.16.2.2 YES manual up up
Loopback1..Loopback5 198.51.100.1 .. 198.51.104.1 up up
```

About the outputs below. The BEFORE and AFTER outputs in Sections 5.2 to 5.6 are the representative outputs given in the Packet Tracer lab guide for this case. Packet Tracer wording can differ slightly. The topology and interface outputs above come from the author's own project files.

5.2 Symptom A: Root Guard blocks a Fab access port

Objective 1. Symptom: SW3 port Fa0/3 is blocked in the root-inconsistent state.

Root cause: SW4's priority was lowered to 0, so it sends a superior BPDU toward SW3. Root Guard on SW3 blocks the port as designed.

- Root Guard protects the intended root. A superior BPDU on a Root Guard port puts the port in a blocking state called root-inconsistent.
- The protection worked correctly; the error is the wrongly lowered priority on SW4. Root Guard must not be removed.

Reproduce (break)

```
SW4(config)# spanning-tree vlan 1 priority 0
```

Observe: BEFORE output (representative)

```
SW3# show spanning-tree inconsistentports
VLAN0001 FastEthernet0/3 Root Inconsistent
Number of inconsistent ports (segment) in the system : 1
%SPANTREE-2-ROOTGUARD_BLOCK: Root guard blocking port
FastEthernet0/3 on VLAN0001.
SW3# show spanning-tree interface fa0/3 detail
SW4# show spanning-tree vlan 1 (claims "This bridge is the root", priority 0)
```

What it shows: Fa0/3 on SW3 is listed as Root Inconsistent and the log shows ROOTGUARD_BLOCK. SW4 claims to be the root with priority 0. That confirms the cause.

Fix

```
SW4(config)# spanning-tree vlan 1 priority 32768
```

Verify: AFTER output (representative)

```
SW3# show spanning-tree inconsistentports
Number of inconsistent ports (segment) in the system : 0
%SPANTREE-2-ROOTGUARD_UNBLOCK: Root guard unblocking port
FastEthernet0/3 on VLAN0001.
```

Verification: No inconsistent ports remain and the UNBLOCK message appears. If the port does not recover, shut and no shut Fa0/3 on SW3, but never remove Root Guard.

5.3 Symptom B: OSPF neighbor stuck in ATTEMPT

Objective 2. Symptom: R1 shows R2 in ATTEMPT while R3 is FULL on the same WAN segment.

Root cause: R2's Gi0/0 was reset to the default BROADCAST network type, while hub R1 uses NON_BROADCAST, so R2 does not answer R1's unicast hellos.

- On NBMA the hub sends unicast hellos to each configured neighbor. ATTEMPT means hellos are being sent and nothing is heard back.
- The other neighbor (R3) is fully adjacent because its side is set up correctly.
- The "missing neighbor statement" version of the fault can be shown too: if R1 has no neighbor 10.0.0.2, R2 never appears at all.

Reproduce (break)

```
R2(config)# interface gi0/0
R2(config-if)# no ip ospf network non-broadcast
R2(config-if)# no ip ospf priority 0
```

Observe: BEFORE output (representative)

```
R1# show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 0 FULL/DROTHER 00:01:41 10.0.0.3 GigabitEthernet0/1
0.0.0.0 0 ATTEMPT/DROTHER 00:01:52 10.0.0.2 GigabitEthernet0/1
R1# show ip ospf interface gi0/1 -> Network Type NON_BROADCAST, Hello 30
R2# show ip ospf interface gi0/0 -> Network Type BROADCAST, Hello 10
```

What it shows: R2 is stuck in ATTEMPT while R3 is FULL, and the two ends of the link show different network types (NON_BROADCAST versus BROADCAST) and hello timers. That is the mismatch.

Fix

```
R2(config)# interface gi0/0
R2(config-if)# ip ospf network non-broadcast
R2(config-if)# ip ospf priority 0
R1(config-router)# neighbor 10.0.0.2 ! add if it was missing on the hub
```

Verify: AFTER output (representative)

```
R1# show ip ospf neighbor
2.2.2.2 0 FULL/DROTHER 00:01:50 10.0.0.2 GigabitEthernet0/1
3.3.3.3 0 FULL/DROTHER 00:01:41 10.0.0.3 GigabitEthernet0/1
```

Verification: Both R2 and R3 reach FULL. Packet Tracer may show Down or Init instead of ATTEMPT; on a real router the state is ATTEMPT.

5.4 Symptom C: EIGRP stub is not advertising a route

Objective 3. Symptom: The Test & Packaging network 192.168.30.0/24 is in R3's own routing table but missing on R2, and R1 cannot ping it.

Root cause: R3 is configured as an EIGRP receive-only stub, which listens but advertises nothing, not even its own connected LAN.

- A stub only advertises what its type allows. The default and "connected summary" types advertise connected and summary routes; receive-only advertises none.
- So the route exists locally on R3, but no neighbor ever learns it.

Reproduce (break)

```
R3(config)# router eigrp 100
R3(config-router)# eigrp stub receive-only
```

Observe: BEFORE output (representative)

```
R3# show ip route 192.168.30.0
C 192.168.30.0/24 is directly connected, Loopback0
R2# show ip route eigrp      -> 192.168.30.0/24 is NOT in the table
R3# show ip protocols       -> EIGRP-IPv4 ... Stub, receive-only
R1# ping 192.168.30.1       -> Request timed out
```

What it shows: The route is present on R3 but absent from R2, and show ip protocols names the stub type as receive-only. The fault is in what R3 advertises, not in the route itself.

Fix

```
R3(config)# router eigrp 100
R3(config-router)# eigrp stub connected summary
```

Verify: AFTER output (representative)

```
R2# show ip route eigrp
D 192.168.30.0/24 [90/130816] via 10.2.0.2, 00:00:12, GigabitEthernet0/2
R1# ping 192.168.30.1      -> !!!!! Success rate is 100 percent
```

Verification: R2 learns the route as D 192.168.30.0/24 and the ping from R1 succeeds.

5.5 Symptom D: eBGP session to the secondary ISP keeps resetting

Objective 4. Symptom: The eBGP session between R1 and ISP2 repeatedly goes down.

Root cause: R1 has a maximum-prefix limit of 3, but ISP2 advertises 5 prefixes. Exceeding the limit tears the session down.

- The limit protects R1 from route leaks and excess routes that could exhaust memory or CPU.
- The session is torn down rather than ignoring the extra routes, because a silently incomplete routing table is unsafe. It stays Idle until cleared or a restart timer expires, and may reset again if the ISP still sends too many prefixes.
- The configured limit is visible with show ip bgp neighbors or show running-config | section bgp; the 5 prefixes can be listed on ISP2 with advertised-routes.

Reproduce (break)

```
R1(config)# router bgp 65100
R1(config-router)# neighbor 172.16.2.2 maximum-prefix 3
```

Observe: BEFORE output (representative)

```
R1# show ip bgp summary
```

```

172.16.2.2 4 65002 11 13 1 0 0 00:00:05 Idle (PfxCt)
%BGP-3-MAXPFX: No. of prefix received from 172.16.2.2 (afi 0) reaches 4, max 3
%BGP-5-ADJCHANGE: neighbor 172.16.2.2 Down - Peer over limit
R1# show ip bgp neighbors 172.16.2.2 | include maximum
Maximum prefixes allowed 3
ISP2# show ip bgp neighbors 172.16.2.1 advertised-routes -> lists the 5 networks

```

What it shows: The MAXPFX message and "Peer over limit" show why the session dropped, and the limit of 3 is below the 5 prefixes ISP2 legitimately sends.

Fix

```

! First confirm with the ISP that the 5 prefixes are legitimate
R1(config-router)# neighbor 172.16.2.2 maximum-prefix 10 80 ! limit 10, warn at 80 percent
R1# clear ip bgp 172.16.2.2

```

Verify: AFTER output (representative)

```

R1# show ip bgp summary
172.16.2.2 4 65002 8 9 6 0 0 00:01:10 5
R1# show ip route bgp -> five B 198.51.10x.0/24 routes

```

Verification: The session is Established and 5 prefixes are received. The limit was raised with headroom and a warning threshold, not removed.

5.6 Symptom E: Burst of large fragmented ICMP echo

Objective 5. Symptom: A capture shows a rapid burst of large, fragmented ICMP echo requests from one source to one internal host.

Root cause: PC2 (192.168.10.66) is flooding Server1 (192.168.20.100) with large echo requests, a ping flood or Ping-of-Death style DoS attempt.

- Pattern: many echo requests larger than the 1500-byte MTU, so they split into fragments, from the same source to the same target, hundreds within a few seconds.
- Likely intent: exhaust the target's CPU, fragment-reassembly memory or bandwidth until the service is slow or unavailable.
- Wireshark filter for the pattern: `icmp.type == 8 && ip.flags.mf == 1` (echo requests with More Fragments set). Packet Tracer Simulation mode with the ICMP filter is used as the capture.

Reproduce (break)

```

PC2> ping -l 10000 -n 200 192.168.20.100 ! use a smaller -l (for example 4000) if Packet
Tracer refuses

```

Observe: BEFORE output (representative)

No.	Source	Destination	Proto	Info
101	192.168.10.66	192.168.20.100	IPv4	Fragmented IP protocol (id=0x2a1f, off=0)
102	192.168.10.66	192.168.20.100	IPv4	Fragmented IP protocol (id=0x2a1f, off=1480)
103	192.168.10.66	192.168.20.100	ICMP	Echo (ping) request (reassembled)

... same source, same destination, hundreds in a few seconds

What it shows: Same source and destination, fragmented large echoes and a very high rate match a flood against a single host.

Fix

```

! Mitigation 1: ACL on R2 (verified in the lab)
R2(config)# ip access-list extended PROTECT-FAB
R2(config-ext-nacl)# deny icmp host 192.168.10.66 host 192.168.20.100 echo
R2(config-ext-nacl)# permit icmp any any echo-reply
R2(config-ext-nacl)# permit ip any any
R2(config)# interface gi0/0
R2(config-if)# ip access-group PROTECT-FAB in

```

```
! Mitigation 2: ICMP rate limit (representative, real IOS)
access-list 120 permit icmp any any echo
class-map match-all ICMP-ECHO / match access-group 120
policy-map LIMIT-ICMP / class ICMP-ECHO / police 64000 conform-action transmit exceed-action drop
interface gi0/0 / service-policy input LIMIT-ICMP
```

Verify: AFTER output (representative)

```
PC2> ping 192.168.20.100      -> Request timed out
PC1> ping 192.168.20.100      -> still works
R2# show access-lists
Extended IP access list PROTECT-FAB
 10 deny icmp host 192.168.10.66 host 192.168.20.100 echo (45 matches)
```

Verification: The ACL match counter rises, PC2 is blocked and PC1 keeps normal connectivity. The rate limit is shown as representative because Packet Tracer usually cannot run it.

6. Results and Conclusion

6.1 Deliverable 1: Root-Cause Analysis Table

Symptom	Likely cause	Evidence	Fix	Verification	Layer
A: Root Guard	SW4 priority 0 sends a superior BPDU	Root Guard block log; inconsistentports lists Fa0/3	Set SW4 priority to 32768	inconsistentports = 0; UNBLOCK log	Layer 2
B: OSPF ATTEMPT	R2 not set to non-broadcast; hub hellos unanswered	R2 in ATTEMPT, R3 FULL; network types differ	Non-broadcast on R2; neighbor 10.0.0.2 on R1	R2 and R3 both FULL	Layer 3
C: EIGRP stub	R3 uses eigrp stub receive-only	Route on R3 not on R2; show ip protocols: receive-only	eigrp stub connected summary	D route on R2; ping works	Layer 3
D: BGP prefix limit	maximum-prefix 3, ISP2 sends 5	MAXPFX log; Idle (PfxCt)	maximum-prefix 10 80; clear ip bgp	Established, 5 prefixes	Internet edge
E: ICMP flood	Large fragmented echo flood from 192.168.10.66	MF fragments, same src and dst, rapid burst	ACL on R2; ICMP rate limit	ACL match counters; PC1 still pings	Host traffic

6.2 Deliverable 2: Are the Symptoms Related? (Objective 6)

Short answer: more than one unrelated fault. Symptoms A and C may come from one maintenance window in the Fab area, but B, D and E are independent.

Taken together, the five symptoms are better explained by several unrelated faults than by one trigger. Each one lives in a different layer and technology: A is a Layer 2 spanning-tree event, B is an OSPF adjacency problem on the WAN, C is a route-advertisement decision in EIGRP, D is a session-protection event at the BGP Internet edge, and E is hostile traffic. A single bad change usually leaves a common fingerprint, such as every protocol on one router failing at the same time, and the symptom log shows nothing like that.

The locations differ too. B involves the HQ and Fab routers across the multipoint WAN, D appears only on the HQ Internet edge, and C is on the Test & Packaging router. Symptoms

A and C, however, both sit in the Fab area and both are described as the result of something done "recently": a switch priority that was lowered and a router that was set up as a stub. Because they share timing and location, they may come from one maintenance window or change ticket, and the change log should be checked. Even so, they are two separate misconfigurations that need two separate fixes. One change cannot explain D, because the BGP edge is at HQ, or E, because a flood is generated by a host and not by a router setting.

Several symptoms also follow a "cause, then protective reaction" pattern. Root Guard blocked the port, the maximum-prefix limit tore down the session, and the receive-only stub withheld its routes, and in each case the feature did exactly what it is designed to do. The real causes sit upstream: a wrongly lowered priority, a prefix limit set below what the ISP legitimately sends, and the wrong stub type. Symptom B is different because the NBMA mismatch is itself the fault, and E is the only symptom with an external actor.

My conclusion is to treat A and C as a possible single change window, to be confirmed from the change log, and to treat B, D and E as independent faults. Each must be fixed and verified on its own, because repairing one will not repair the others. To prevent repeats, the network team should use change control, Root Guard together with BPDU Guard on edge ports, prefix limits with headroom and warning thresholds, a review of EIGRP stub design, and ICMP rate limiting and ACLs at the Fab boundary.

(Narrative length: 404 words.)

6.3 Deliverable 3: Prioritized Remediation Plan (Objective 7)

Order	Symptom	Business impact	Why this order	Verification
1	B: OSPF WAN	HQ to Fab connectivity on the backbone is degraded; production systems and ERP depend on it	The WAN is the backbone; other sites and fixes rely on it	show ip ospf neighbor (all FULL)
2	C: EIGRP stub	Test & Packaging network is unreachable from the rest of the company; test results cannot reach HQ	A whole site is cut off from some paths	show ip route eigrp; ping 192.168.30.1
3	A: Root Guard	Devices behind the blocked Fab port lose network; STP security risk	One access port, local impact, quick fix	show spanning-tree inconsistentports
4	E: ICMP flood	Possible slowdown of a Fab host; availability risk but no outage yet	Contain quickly: the ACL takes minutes and can be done in parallel	show access-lists (match counters)
5	D: BGP secondary	Only the backup Internet path is flapping; the primary ISP still carries traffic	Redundancy is lost, not an outage, so it goes last	show ip bgp summary (Established)

Quick win: the ICMP ACL for symptom E takes about two minutes and can be applied first as containment while B, C and A are fixed.

6.4 Findings

- All five symptoms were reproduced, diagnosed from show-command evidence, fixed with a targeted change and verified with the same command.

- In three cases (A, D and C) the protection or feature behaved correctly: Root Guard, the maximum-prefix limit and the stub each did what they were designed to do. The causes were the settings around them.
- B is the only fault where the configuration mismatch itself broke the adjacency, and E is the only one with an external actor.
- The faults are not one incident. Fixing one does not fix the others, so each needs its own verification.

6.5 Limitations

- The work is simulated in Cisco Packet Tracer, which does not provide the full feature set of real Cisco IOS.
- The BEFORE and AFTER outputs in this report are the lab guide's representative outputs. They should be replaced with the author's own Packet Tracer screenshots for each fault where the assessor requires real captures.
- Wireshark is not part of Packet Tracer, so Simulation mode stands in; the Wireshark filter is given as representative.
- ICMP rate limiting (policing) is shown as representative configuration. The ACL is the lab-verified mitigation.
- Packet Tracer may show Down or Init instead of ATTEMPT for symptom B, and may refuse very large ping sizes (use a smaller -l value so the packets still fragment).
- ISP1 is left out of the lab; only the secondary ISP (ISP2) is simulated.

6.6 Recommendations to Prevent Repeats

- Change control: review the change log, since A and C may share one change window.
- Use Root Guard together with BPDU Guard on edge ports.
- Set BGP prefix limits with headroom and a warning threshold, and confirm unexpected prefixes with the ISP.
- Review EIGRP stub design so each stub type matches what the site must advertise.
- Apply ICMP rate limiting and ACLs at the Fab boundary, and filter or drop fragments to servers where possible.

6.7 Conclusion

The Coppervale outage was not one failure but five separate ones, each at a different layer of the network. A systematic Reproduce, Observe, Explain, Fix, Verify workflow isolated each cause from evidence rather than guesswork, and each fix was proven with the same command that exposed the fault. Ranking the repairs by business impact (OSPF WAN first, BGP backup last, with the ICMP ACL applied early as quick containment) restores the most important connectivity first. The project shows how routing, switching and security features each fail safely, and why the root cause usually lies in the configuration around the protection.

7. References

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